

MODULE 10.2

States of Data

Endpoint Security | Cisco Networking Academy

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Cybersecurity Essentials

Topics Covered:

10.2 States of Data

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10.2 States of Data

Three States Where Data Must Be Protected

10.2

Understanding Data States

Data exists in three different states: at rest (stored), in transit (moving across a network), and in process (actively being used). Each state presents unique security challenges that require tailored countermeasures.

DATA AT REST

Information stored on physical or cloud media (HDDs, SSDs, USB drives, databases). Must be encrypted and access-controlled at all times.

DATA IN TRANSIT

Information moving between systems over networks. Must be protected with TLS/SSL, VPNs, or IPsec to prevent interception and tampering.

DATA IN PROCESS

Information actively being used in RAM or CPU operations. Vulnerable to memory scraping, buffer overflows, and cold-boot attacks.

10.2.2 Data at Rest

Securing Stored Data

10.2.2

What Is Data at Rest?

Data at rest refers to inactive data stored physically on any digital form (e.g., hard disks, solid-state drives, USB flash drives, cloud storage, and databases). It must be protected by encryption, access control, and physical security measures.

Full-Disk Encryption

Encrypt entire storage volumes using AES-256, BitLocker (Windows), or FileVault (macOS) to render data unreadable if a device is stolen.

File-Level Encryption

Encrypt individual files or folders so only authorised users with the correct key can read the data, even if the disk is accessed directly.

Access Controls & Permissions

Apply least-privilege principles using ACLs and RBAC so that only authorised roles can read, write, or delete stored data.

Physical Security

Secure server rooms, data centres, and storage devices with locks, cameras, and asset-tracking systems to prevent theft.

Encrypted Backups

Ensure all backup copies are encrypted and stored securely, with regularly tested restore procedures to confirm integrity.

10.2.3 Challenges of Protecting Stored Data

Vulnerabilities in Data at Rest

10.2.3

Why Is Stored Data Difficult to Protect?

Even with encryption in place, stored data faces a wide range of threats. Understanding these challenges helps organisations design layered defences that address both technical and human factors.

Physical Theft

Lost or stolen laptops, USB drives, and hard disks expose unencrypted data directly. Encryption must be enabled before the device leaves a secure environment.

Insider Threats

Authorised users may intentionally or accidentally leak, copy, or destroy sensitive files. DLP tools, audit logs, and least-privilege access limit this risk.

Weak or Missing Encryption

Outdated algorithms (DES, MD5), poor key management, or skipped encryption leaves data vulnerable. Regular audits enforce cryptographic standards.

Unmanaged and Shadow IT

Personal devices (BYOD) and unsanctioned cloud services create uncontrolled data stores outside IT oversight, making policy enforcement difficult.

Data Sprawl

Duplicate copies spread across multiple locations are difficult to track, update, and securely delete when no longer needed.

10.2.5 Methods of Transmitting Data

How Data Moves Between Systems

10.2.5

Transmission Methods and Associated Risks

Data in transit travels via several physical and logical channels, each with different security implications. Selecting the right transmission method and applying appropriate controls is essential to protect data integrity and confidentiality.

Step 1

Wired (Ethernet / Fibre)

High-speed and physically contained. Easier to secure than wireless but vulnerable to physical tapping. Use network segmentation and 802.1X authentication.

Step 2

Wireless (Wi-Fi /

Flexible but transmits data through the air. Requires WPA3 encryption, hidden SSIDs, and device authentication to reduce eavesdropping risk.

Step 3

Network Protocols

TCP/IP, HTTPS, SFTP, and SSH provide structured, encrypted communication. Avoid legacy plaintext protocols (FTP, Telnet, HTTP) in sensitive contexts.

Step 4

Physical Transfer Media

USB drives, optical discs, and removable media carry data outside the network. Enforce encryption, disable autorun, and track media with DLP policies.

10.2.6 Challenges of Data in Transit

Securing Data as It Moves

10.2.6

Why Is Data in Transit Vulnerable?

Data moving across networks is exposed to a range of interception and manipulation attacks. Without appropriate protection, sensitive information can be captured, altered, or replayed by malicious actors anywhere along the transmission path.

Eavesdropping / Sniffing

Attackers use packet-capture tools (e.g., Wireshark) to passively read unencrypted traffic on shared or compromised network segments.

Man-in-the-Middle (MitM)

An attacker intercepts communication between two parties, reading or modifying data in real time without either party's knowledge.

Replay Attacks

Captured authentication tokens or commands are retransmitted to repeat valid operations (e.g., re-authorising a transaction) after the original session ends.

Weak / Legacy Protocols

Older protocols such as FTP, Telnet, and HTTP transmit data in plaintext. Replace with SFTP, SSH, HTTPS, and TLS 1.3 to ensure encryption in transit.

Countermeasures

TLS/SSL, IPsec VPNs, certificate pinning, HSTS, mutual TLS (mTLS), and network-layer monitoring collectively defend data in transit from interception.

10.2.7 Data in Process

Protecting Data During Active Use

10.2.7

What Is Data in Process?

Data in process is data being actively created, updated, calculated, or transmitted. It resides temporarily in volatile memory (RAM) and CPU registers, making it uniquely vulnerable to memory-based attacks that bypass traditional encryption.

Definition

Any data currently loaded into RAM, CPU caches, or application buffers, including opened files, active sessions, calculations, and database queries.

Cold-Boot and RAM Attacks

Physical access to a running or recently powered-off machine can allow extraction of encryption keys and sensitive data from volatile memory.

Buffer Overflow Exploits

Writing more data than a buffer can hold overwrites adjacent memory, potentially giving attackers code-execution capability or access to sensitive values.

Secure Coding Practices

Input validation, bounds checking, memory-safe languages (Rust, Go), and compiler protections (ASLR, stack canaries) reduce in-process vulnerabilities.

Trusted Execution Environments

Hardware-isolated enclaves (Intel SGX, ARM TrustZone) execute sensitive computations in a protected memory region, shielded from the OS and hypervisor.

10.2.9 Oh No! A Security Breach...

Case Study: What Happens When Protection Fails

10.2.9

Anatomy of a Data Breach and Incident Response

A security breach occurs when an attacker successfully bypasses controls protecting data at rest, in transit, or in process. Understanding the typical breach lifecycle helps organisations respond effectively and minimise damage.

1. Identification

The breach is detected via IDS/IPS alerts, SIEM correlation, anti-malware tools, unusual login activity, or an external tip. Time-to-detect is critical.

2. Containment

Isolate compromised systems immediately, disable accounts, segment the network, and block external connections to prevent lateral movement and further data loss.

3. Eradication

Remove malware, close exploited vulnerabilities, revoke compromised credentials, and patch affected systems before returning them to service.

4. Recovery

Restore clean data from verified, encrypted backups. Validate integrity using hashes. Gradually bring systems back online with enhanced monitoring.

5. Lessons Learned

Conduct a post-incident review. Update policies, retrain staff, improve detection rules, and remediate root causes to prevent recurrence.

Module 10.2 — Summary

10.2 States of Data

Data exists in three states: at rest, in transit, and in process, each needing dedicated controls.

10.2.3 Challenges

Theft, insider threats, weak encryption, shadow IT, and data sprawl complicate protection.

10.2.6 Data in Transit

Eavesdropping, MitM, replay, and legacy protocols threaten moving data; TLS and VPNs protect it.

10.2.9 Security Breach

Breaches follow identify-contain-eradicate-recover; lessons learned drive ongoing improvement.

10.2.2 Data at Rest

Encrypt stored data with AES-256; enforce access controls and physical security.

10.2.5 Transmission

Wired, wireless, and protocol-based channels each carry risk; use encrypted protocols.

10.2.7 Data in Process

RAM-based attacks and buffer overflows target active data; secure coding and TEEs defend it.